fm-ggp2

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1/6/2023

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Preface

Welcome! This website contains a collection of graphs created using `ggplot2` (and friends!).

Install `ggplot2` using the code below:

```
install.packages("ggplot2")
library(ggplot2)
```

Or you can install `ggplot2` as part of the `tidyverse`:

```
install.packages("tidyverse")
library(tidyverse)
```

The Data

To improve reproducibility, the majority of the graphs are built using the `palmerpenguins::penguins` data.

...so...many...PENGUINS!

Figure 1: Artwork by allison_horst

```
install.packages("palmerpenguins")
library(palmerpenguins)
```

Some of the graphs use datasets from the `fivethirtyeight` package.

```
install.packages("fivethirtyeight")
library(fivethirtyeight)
```

Expand to view the data in the `fivethirtyeight` package

To view a table of available datasets in the `fivethirtyeight` package, view the `Data Frame Name` and `Article Title` columns in the `datasets_master` table:

Data Frame Name	Article Title
ahca_polls	Why The GOP Is So Hell-Bent On Passing An Unpopular Health Care Bill
airline_safety	Should Travelers Avoid Flying Airlines That Have Had Crashes in the Past?
antiquities_act	Trump Might Be The First President To Scrap A National Monument
august_senate_polls	How Much Trouble Is Ted Cruz Really In?
avengers	Joining The Avengers Is As Deadly As Jumping Off A Four-Story Building

1–5 of 148 rows

Previous 1 2 3 4 5 ... 30 Next

A *few* of the graphs are built using the `ggplot2movies::movies` data.

```
install.packages("ggplot2movies")
library(ggplot2movies)
```

The Graphs

The gallery follows a Rule of Least Power Principle, in the sense that "*a language with a straightforward syntax may be easier to analyze than an otherwise equivalent one with more complex structure.*"

In other words, assuming the reader has some understanding of R and the `tidyverse`, the code for each graph is meant to be read and understood *without* having to run it.

Graph Categories

Graphs have been categorized into the following types:

- Univariate graphs
  - Amounts
  - Proportions
- Comparing Distributions
- Relationships

Some graphs would justifiably belong in more than one category, and wherever this is the case, I've tried to include links to other uses in the notes.

Code style & structure

Each graph has, at minimum, the same two sections and four tabs:

1. **Packages** and **Data**
   - Code for installing development version of packages (if necessary) for graphs *and* data
   - Any steps used to create (i.e., manipulate and prepare) the data for the graph
2. **Code** and **Graph**
   - Code to build the labels and graph layers
     - *Graph labels have the* `labs_` *prefix*
     - *Graph layers have a* `ggp2_` *prefix*

Expand to view the graph code structure

Packages

PACKAGES:
Install packages.

```
library(ggplot2)
```

Data

DATA:
Description of data

```
df <- tibble::tibble(X = sample(x = 1:100, 10, FALSE),
                     Y = rlnorm(10, 1, 3))
```

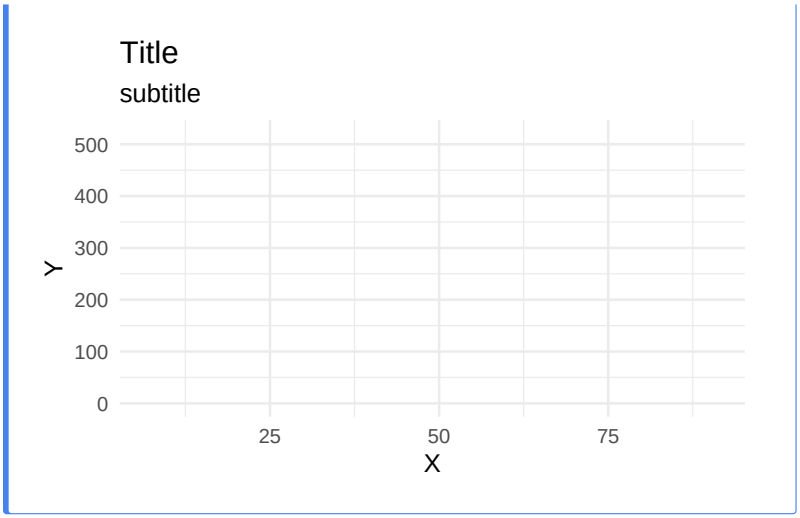
Code

CODE:
Code for creating graph

```
# labels
labs_graph <- ggplot2::labs(title = "Title",
                            subtitle = "subtitle",
                            x = "X", y = "Y")
# layers
ggp2_graph <- ggplot2::ggplot(data = df,
    mapping = aes(x = X, y = Y)) +
    ggplot2::geom_blank()
# graph
ggp2_graph +
    labs_graph
```

Graph

GRAPH:



Code Style:

I've attempted to balance brevity and clarity, but with the assumption that its best to err on the latter. I've also followed the general principle that if a graph can be *easily* built using one of `ggplot2` 's `geom_*` functions, that method is shown first.

Extensions

Below are the graphs requiring additional packages/extensions:

- ☒ Waffle charts
  - `ggwaffle` package

```
# Waffle Charts
devtools::install_github("liamgilbey/ggwaffle")
library(ggwaffle)
```

- ☒ Pie Charts
  - `ggpubr` package

```
# Pie Charts
devtools::install_github("kassambara/ggpubr")
library(ggpubr)
```

☒ Mosaic Plots

- `ggmosaic` package

```
# Mosaic Charts
devtools::install_github("haleyjeppson/ggmosaic")
library(ggmosaic)
```

☒ Treemaps

- `treemapify` package

```
# Treemaps
devtools::install_github("wilkox/treemapify")
library(treemapify)
```

☒ Bee-swarm Plots

- `ggbeeswarm` package

```
# Bee-swarm Plots
devtools::install_github("eclarke/ggbeeswarm")
library(ggbeeswarm)
```

☒ Ridgeline Plots

- `ggridges` package

```
# Ridgeline plots
devtools::install_github("wilkelab/ggridges")
library(ggridges)
```

☒ Rain-cloud plots

- `raincloudplots` and `ggdist` packages

```
# Rain-cloud plots
remotes::install_github('jorvlan/raincloudplots')
remotes::install_github('mjskay/ggdist')
library(raincloudplots)
library(ggdist)
```

- ☒ Alluvial charts
  - ggalluvial package

```
# Alluvial charts
devtools::install_github("corybrunson/ggalluvial")
library(ggalluvial)
```

- ☒ Bump charts
  - ggbump package

```
# Bump charts
devtools::install_github("davidsjoberg/ggbump")
library(ggbump)
```

- ☒ Parallel Sets
  - ggforce package

```
# Bump charts
devtools::install_github("thomasp85/ggforce")
library(ggforce)
```

- ☒ Stream Plots
  - ggstream package

```
# Stream plots
devtools::install_github("davidsjoberg/ggstream")
library(ggstream)
```

Theme

The theme used in the graphs is custom and uses combined elements from `ggplot2::theme_minimal()` and `ggplot2::theme_void()`. View it here.

Part I

Univariate